

# 廖振宇

## 工作经历

- 2021-至今 副研究员, 华中科技大学, 电子信息与通信学院  
2020-2021 博士后研究员, 加州大学伯克利分校, 统计系, 合作导师: Michael Mahoney 教授

## 教育经历

- 2016-2019 博士, 巴黎萨克雷大学, 数学与计算机, 合作导师: Romain Couillet 教授  
2014-2016 硕士, 巴黎萨克雷大学, 信号与图像处理, 合作导师: Romain Couillet 教授  
2010-2014 本科, 华中科技大学, 光电信息工程

## 科研获奖

- 2026 Springer Nature Editor of Distinction Award of *Author Service Award 2026* (作为 Statistics and Computing 编委)  
2025 IEEE Senior Member  
2025 **MATRIX-Simons Travel Grant**, MATRIX, Creswick, 澳大利亚  
2025 加拿大 CRM-Simons 访问教授, 蒙特利尔大学 **Centre de recherches mathématiques** (签证原因未成行)  
2024 法国 ANR-LabEx-CIMI 访问教授, 图卢兹第三大学 **Centre International de Mathématiques et Informatique de Toulouse (CIMI)**  
2023 湖北省人才计划  
2021 武汉市人才计划  
2021 华中科技大学东湖青年学者  
2019 巴黎萨克雷大学 **ED STIC 优秀博士论文**  
2016 巴黎萨克雷大学 Supélec Foundation Ph.D. Fellowship

## 学术专著

- CUP 2022** Romain Couillet and **Zhenyu Liao** (2022). Random Matrix Methods for Machine Learning. Cambridge University Press. DOI: [10.1017/9781009128490](https://doi.org/10.1017/9781009128490).

## 代表论文

### 会议

- ICASSP 2026** Yue Xu and **Zhenyu Liao** (2026). An Improved Convergence Analysis of Gossip Methods for Large Random Graphs. *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 22562–22566. DOI: [10.1109/ICASSP55912.2026.11461477](https://doi.org/10.1109/ICASSP55912.2026.11461477).
- ICASSP 2026** Yessin Moahker, Malik Tiomoko, Cosme Louart, and **Zhenyu Liao** (2026). A Random Matrix Perspective of Echo State Networks: From Precise Bias-Variance Characterization to Optimal Regularization. *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 20716–20720. DOI: [10.1109/ICASSP55912.2026.11464627](https://doi.org/10.1109/ICASSP55912.2026.11464627).
- SIGMOD 2026** Jiuqi Wei, **Zhenyu Liao**, Ruoyu Han, Quanqing Xu, Chuanhui Yang, and Themis Palpanas (May 2026). TaCo: Data-adaptive and Query-aware Subspace Collision for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (SIGMOD)* 4(3). DOI: [10.1145/3802118](https://doi.org/10.1145/3802118). URL: <https://doi.org/10.1145/3802118>.
- ICML 2025** Chengmei Niu, **Zhenyu Liao**, Zenan Ling, and Michael W. Mahoney (13–19 Jul 2025). Fundamental Bias in Inverting Random Sampling Matrices with Application to Sub-sampled Newton. *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. Vol. 267. Proceedings of Machine Learning Research. PMLR, pp. 46649–46692. URL: <https://proceedings.mlr.press/v267/niu25c.html>.

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更新时间: 2026 年 6 月



- ICLR 2025** Xiaoyi Mai and **Zhenyu Liao** (2025). The Breakdown of Gaussian Universality in Classification of High-dimensional Mixtures. *International Conference on Learning Representations (ICLR)*. URL: <https://openreview.net/forum?id=UrKbn51HjA>.
- SIGMOD 2025** Jiuqi Wei, Xiaodong Lee, **Zhenyu Liao**, Themis Palpanas, and Botao Peng (Feb. 2025). Subspace Collision: An Efficient and Accurate Framework for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (SIGMOD)* **3**(1). DOI: [10.1145/3709729](https://doi.org/10.1145/3709729). URL: <https://doi.org/10.1145/3709729>.
- EUSIPCO 2024** Wei Yang, Zhengyu Wang, Xiaoyi Mai, Zenan Ling, Robert Caiming Qiu, and **Zhenyu Liao** (Aug. 2024). Inconsistency of ESPRIT DoA Estimation for Large Arrays and a Correction via RMT. *2024 32nd European Signal Processing Conference (EUSIPCO)*, pp. 2722–2726. DOI: [10.23919/EUSIPCO63174.2024.10715081](https://doi.org/10.23919/EUSIPCO63174.2024.10715081).
- ICML 2024** Zenan Ling, Longbo Li, Zhanbo Feng, Yixuan Zhang, Feng Zhou, Robert C. Qiu, and **Zhenyu Liao** (21–27 Jul 2024). Deep Equilibrium Models Are Almost Equivalent to Not-so-deep Explicit Models for High-dimensional Gaussian Mixtures. *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. 235. PMLR, pp. 30585–30609. URL: <https://proceedings.mlr.press/v235/ling24a.html>.
- NeurIPS 2022** Lingyu Gu, Yongqi Du, Yuan Zhang, Di Xie, Shiliang Pu, Robert Qiu, and **Zhenyu Liao** (2022). "Lossless" Compression of Deep Neural Networks: A High-dimensional Neural Tangent Kernel Approach. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 35. Curran Associates, Inc., pp. 3774–3787. URL: [https://proceedings.neurips.cc/paper\\_files/paper/2022/hash/185087ea328b4f03ea8fd0c8aa96f747-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2022/hash/185087ea328b4f03ea8fd0c8aa96f747-Abstract-Conference.html).
- ICLR 2022** Hafiz Tiomoko Ali, **Zhenyu Liao**, and Romain Couillet (2022). Random matrices in service of ML footprint: ternary random features with no performance loss. *International Conference on Learning Representations (ICLR)*. URL: <https://openreview.net/forum?id=qwULHx9zld>.
- NeurIPS 2021** **Zhenyu Liao** and Michael W Mahoney (2021). Hessian Eigenspectra of More Realistic Nonlinear Models. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 34. Curran Associates, Inc., pp. 20104–20117. URL: <https://papers.nips.cc/paper/2021/hash/a7d8ae4569120b5bec12e7b6e9648b86-Abstract.html>.
- COLT 2021** Michal Dereziński, **Zhenyu Liao**, Edgar Dobriban, and Michael Mahoney (Aug. 2021). Sparse sketches with small inversion bias. *Proceedings of Thirty Fourth Conference on Learning Theory (COLT)*. Vol. 134. PMLR, pp. 1467–1510. URL: <https://proceedings.mlr.press/v134/dereziński21a.html>.
- AISTATS 2021** Fanghui Liu, **Zhenyu Liao**, and Johan Suykens (Apr. 2021). Kernel Regression in High Dimension: Refined Analysis beyond Double Descent. *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*. Vol. 130. PMLR, pp. 649–657. URL: <http://proceedings.mlr.press/v130/liu21b.html>.
- ICLR 2021** **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney (2021a). Sparse Quantized Spectral Clustering. *International Conference on Learning Representations (ICLR)*. URL: <https://openreview.net/forum?id=pBqLS-7KYAF>.
- NeurIPS 2020** Michal Dereziński, Feynman T Liang, **Zhenyu Liao**, and Michael W. Mahoney (2020). Precise expressions for random projections: Low-rank approximation and randomized Newton. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. Curran Associates, Inc., pp. 18272–18283. URL: <https://papers.nips.cc/paper/2020/hash/d40d35b3063c11244fbf38e9b55074be-Abstract.html>.
- NeurIPS 2020** **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney (2020). A Random Matrix Analysis of Random Fourier Features: Beyond the Gaussian Kernel, A Precise Phase Transition, and the Corresponding Double Descent. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. Curran Associates, Inc., pp. 13939–13950. URL: <https://papers.nips.cc/paper/2020/hash/a03fa30821986dff10fc66647c84c9c3-Abstract.html>.
- ICASSP 2019** Xiaoyi Mai, **Zhenyu Liao**, and Romain Couillet (May 2019). A Large Scale Analysis of Logistic Regression: Asymptotic Performance and New Insights. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, pp. 3357–3361. DOI: [10.1109/ICASSP.2019.8683376](https://doi.org/10.1109/ICASSP.2019.8683376).
- EUSIPCO 2018** Romain Couillet, **Zhenyu Liao**, and Xiaoyi Mai (Sept. 2018). Classification Asymptotics in the Random Matrix Regime. *The 26th European Signal Processing Conference (EUSIPCO)*. IEEE, pp. 1875–1879. DOI: [10.23919/EUSIPCO.2018.8553034](https://doi.org/10.23919/EUSIPCO.2018.8553034).
- ICML 2018** **Zhenyu Liao** and Romain Couillet (July 2018a). The Dynamics of Learning: A Random Matrix Approach. *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80. PMLR, pp. 3072–3081. URL: <http://proceedings.mlr.press/v80/liao18b.html>.
- ICML 2018** **Zhenyu Liao** and Romain Couillet (July 2018b). On the Spectrum of Random Features Maps of High Dimensional Data. *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80. PMLR, pp. 3063–3071. URL: <http://proceedings.mlr.press/v80/liao18a.html>.
- ICASSP 2017** **Zhenyu Liao** and Romain Couillet (Mar. 2017). Random Matrices Meet Machine Learning: A Large Dimensional Analysis of LS-SVM. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, pp. 2397–2401. DOI: [10.1109/ICASSP.2017.7952586](https://doi.org/10.1109/ICASSP.2017.7952586).

## 期刊

- SPM 2026** **Zhenyu Liao** and Michael W. Mahoney (2026). Random Matrix Theory for Deep Learning: Beyond Eigenvalues of Linear Models. *IEEE Signal Processing Magazine* **43**(2), 93–106. DOI: [10.1109/MSP.2025.3618012](https://doi.org/10.1109/MSP.2025.3618012).

- TSP 2026** Zhengyu Wang, Wei Yang, Xiaoyi Mai, Zenan Ling, **Zhenyu Liao**, and Robert C. Qiu (2026). A Large-Dimensional Analysis of ESPRIT DoA Estimation: Inconsistency and a Correction via RMT. *IEEE Transactions on Signal Processing* **74**, 2290–2303. DOI: [10.1109/TSP.2026.3683673](https://doi.org/10.1109/TSP.2026.3683673).
- TKDE 2024** Zhanbo Feng, Yuanjie Wang, Jie Li, Fan Yang, Jiong Lou, Tiebin Mi, Robert. C. Qiu, and **Zhenyu Liao** (2025). Robust and Communication-Efficient Federated Domain Adaptation via Random Features. *IEEE Transactions on Knowledge and Data Engineering* **37**(3), 1411–1424. DOI: [10.1109/TKDE.2024.3510296](https://doi.org/10.1109/TKDE.2024.3510296).
- PRA 2023** Jingcheng Wang, Shaoliang Zhang, Jianming Cai, **Zhenyu Liao**, Christian Arenz, and Ralf Betzholz (Aug. 2023). Robustness of random-control quantum-state tomography. *Phys. Rev. A* **108** (2), 022408. DOI: [10.1103/PhysRevA.108.022408](https://doi.org/10.1103/PhysRevA.108.022408). URL: <https://link.aps.org/doi/10.1103/PhysRevA.108.022408>.
- MCRF 2022** Yacine Chitour, **Zhenyu Liao**, and Romain Couillet (2023). A geometric approach of gradient descent algorithms in linear neural networks. *Mathematical Control and Related Fields* **13**(3), 918–945. DOI: [10.3934/mcrf.2022021](https://doi.org/10.3934/mcrf.2022021).
- JSTAT 2021** **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney (Dec. 2021b). A random matrix analysis of random Fourier features: beyond the Gaussian kernel, a precise phase transition, and the corresponding double descent. *Journal of Statistical Mechanics: Theory and Experiment* **2021**(12), 124006. DOI: [10.1088/1742-5468/ac3a77](https://doi.org/10.1088/1742-5468/ac3a77). URL: <https://doi.org/10.1088/1742-5468/ac3a77>.
- TSP 2019** **Zhenyu Liao** and Romain Couillet (Feb. 2019b). A Large Dimensional Analysis of Least Squares Support Vector Machines. *IEEE Transactions on Signal Processing* **67**(4), 1065–1074. DOI: [10.1109/TSP.2018.2889954](https://doi.org/10.1109/TSP.2018.2889954).
- AAP 2018** Cosme Louart, **Zhenyu Liao**, and Romain Couillet (Apr. 2018). A Random Matrix Approach to Neural Networks. *The Annals of Applied Probability* **28**(2), 1190–1248. DOI: [10.1214/17-AAP1328](https://doi.org/10.1214/17-AAP1328). URL: <https://doi.org/10.1214/17-AAP1328>.

## 科研项目

### 在研

- 2026-2029 国家自然科学基金面上项目：基于随机矩阵方法的 *Transformer* 大模型压缩：从缩放定律极限理论到动态自适应模型压缩方法 (NSFC-12571561)，44 万元，**主持**；Co-PI: **Cosme Louart**, SDS, 香港中文大学（深圳）
- 2026-2028 华中科技大学“基础研究支持计划”数学专项项目：基于随机矩阵方法的大规模机器学习理论和方法 (2025BRAXB0004)，38 万元，**主持**
- 2025-2030 国家重点研发计划青年科学家项目：揭示大模型机理的机器学习基础理论研究 (2025YFA1018600)，100/300 万元，项目骨干；PI: 束俊，数学与统计学院，西安交通大学
- 2024-2026 华为技术有限公司校企合作项目： *Algorithm and Theory of High-dimensional Dynamical Systems* (TC20240702035)，80 万元，**主持**；Co-PI: 凌泽南，华中科技大学电信学院

### 已结题

- 2024-2026 广东省人工智能数理基础重点实验室开放基金：基于随机矩阵方法的 *Transformer* 模型泛化理论研究 (OFA00003)，10 万元，**主持**；Co-PI: **Jeff Yao**, SDS, 香港中文大学（深圳）
- 2023-2025 国家自然科学基金青年科学基金项目：基于随机矩阵方法的神经网络模型剪枝基础理论研究 (NSFC-62206101)，30 万元，**主持**
- 2023-2025 华为技术有限公司校企合作项目：随机矩阵理论驱动的通信理论和算法研究 (TC20231122043)，59 万元，**主持**
- 2022-2025 国家自然科学基金“面向未来通信的数学基础（信息论）”专项项目：智能反射面辅助的新型无线通信数学理论与数学技术 (NSFC-12141107)，300 万元，项目骨干；PI: 邱才明，华中科技大学电信学院
- 2021-2022 中国计算机学会 CCF-海康威视斑头雁基金项目：基于随机矩阵和信息瓶颈理论的神经网络表达和压缩的研究 (20210008)，28 万元，**主持**；Co-PI: 万凯，华中科技大学电信学院
- 2018-2021 NSF Research Grant, *Combining Stochastics and Numerics for Improved Scalable Matrix Computations* (NSF-1815054)，500k 美元，项目骨干；PI: Michael W. Mahoney, UC Berkeley, USA
- 2018-2021 法国高等教育、研究与创新部：GSTATS-IDEX DataScience Chair, 300k 欧元，项目骨干；PI: Romain Couillet, U Grenoble Alpes, France
- 2015-2017 法国自然科学基金委： *Random Matrix Theory for Large Dimensional Graphs* (ANR-14-CE28-0006)，300k 欧元，项目骨干；PI: Romain Couillet, U Paris-Saclay, France

## 教学改革项目

2024-2025 华中科技大学国家卓越工程师学院 2025 年“卓越工程师自主培养新范式”项目：产教融合卓越工程师国际化视野培养，3 万元，**主持**（结题）

## 科研服务

- 学会兼职** 中国现场统计研究会随机矩阵理论与应用分会副秘书长，中国现场统计研究会大数据统计分会理事，中国商业统计学会人工智能分会理事，中国通信学会短距无线通信委员会秘书
- 评审专家** 欧洲研究理事会 **ERC**，加拿大自然科学与工程研究委员会 **NSERC**，以色列科学基金会 **ISF**，中国国家自然科学基金委员会 **NSFC**
- 国际会议领域主席** **ICML** 2025–2026, **NeurIPS** 2025–2026, **ICLR** 2025–2026, **AISTATS** 2026, **IJCNN** 2025–2026
- 国际期刊编委** Springer **Statistics and Computing** (2025 年至今)
- 国际会议审稿人** **ICLR**, **ICML**, **NeurIPS**, **SODA**, **AISTATS**, **AAAI**, **ECAI**, **IEEE-ICASSP**, **IEEE-CAMSAP**.
- 数学与数据科学期刊审稿人** **SIAM Review (SIREV)**, **Foundations of Computational Mathematics (FoCM)**, **Physical Review X (PRX)**, **Journal of Mathematical Physics**, **Springer Statistics and Computing (STCO)**, **SIAM Journal on Scientific Computing (SISC)**, **Random Matrices: Theory and Applications (RMTA)**, **Latin American Journal of Probability and Mathematical Statistics (ALEA)**, **Annals of Applied Probability (AAP)**, **Electronic Communications in Probability (ECP)** **Electronic Journal of Statistics (EJS)**, **Journal of Applied Mathematics and Computing (JAMC)**, **AIMS Mathematics**.
- 机器学习和 AI 期刊审稿人** **Journal of Machine Learning Research (JMLR)**, **IEEE Trans. on Pattern Analysis and Machine Intelligence (IEEE-TPAMI)**, **IEEE Signal Processing Letters (SPL)**, **IEEE Trans. on Signal Processing (IEEE-TSP)**, **IEEE Trans. on Multimedia (IEEE-TMM)**, **Engineering Applications of Artificial Intelligence (EAAI)**, **Frontiers of Computer Science**, **Neural Networks (NEUNET)**, **IEEE Trans. on Neural Networks and Learning Systems (IEEE-TNNLS)**, **IEEE Transactions on Industrial Informatics (IEEE-TII)**, **Transactions on Machine Learning Research (TMLR)**, **Pattern Recognition (PR)**, **Neural Processing Letters (NPL)**.
- 学术活动**
1. **CSIAM** 第七届大数据与人工智能科学大会（**CSIAM-BDAI 2026**）TM10 大模型的“理论指导实践”，2026 年 7 月 17 至 7 月 19 日，福建厦门
    - 组织委员会：刘方辉、吴磊、廖振宇
    - 报告人：陆一平、刘勇、徐志钦、吕凯风、方聪、束俊、孙若愚
  2. **2026 中国机器学习与科学应用大会（CSML 2026）** 现代机器学习理论与方法，2026 年 8 月 7 日至 8 月 9 日，上海
    - 组织委员会：束俊、廖振宇
    - 报告人：束俊、张慧铭、邹荻凡、张俊杰
  3. “Universality and Dynamics in High-Dimensional Learning and Inference” Workshop at **2026 IEEE International Symposium on Information Theory (ISIT 2026)**，2026 年 7 月 3 日，广州
    - 组织委员会：**Rishabh Dudeja**，廖振宇，马俊杰，**Arian Maleki**
  4. **2025 年大模型数理统计基础研讨会**，2025 年 7 月，吉林省长白山
    - 组织委员会：方聪，廖振宇，杨朋昆
    - 在线直播吸引超过 3.5 万观众
  5. 中国现场统计研究会随机矩阵理论与应用分会成立大会，2025 年 1 月，吉林省长春市
    - 组织委员会：姚建峰，郑术蓉，胡江，姜丹丹，廖振宇
  6. **General co-chair of 10th EAI International Conference on IoT as a Service (EAI IoTaaS 2024)**, Wuhan, China, 2024
  7. 华中科技大学-巴黎萨克雷大学 2022 联合研讨会“数据科学中的数学奥秘”
    - 召集人：廖振宇，Yacine Chitour
    - 吸引超过 4 万名相关领域的科研人员、老师和同学参与
    - 线上回放链接：<https://www.bilibili.com/video/BV1G8411b7ir/>
  8. 1st Workshop in High-dimensional Learning Dynamics (**HiLD**) at **ICML 2023**, Honolulu, Hawaii
    - 组织委员会：Mihai Nica, Courtney Paquette, Elliot Paquette, Andrew Saxe, and René Vidal
    - 报告嘉宾：**Sanjeev Arora**, **SueYeon Chung**, **Murat A. Erdogdu**, **Surya Ganguli**, and **Andrea Montanari**

## 推荐人



**邱才明** 华中科技大学电子信息与通信学院院长, IEEE Fellow。Email: [caiming@hust.edu.cn](mailto:caiming@hust.edu.cn)

**Michael Mahoney** Associate Adjunct Professor at Department of Statistics, UC Berkeley, CA, USA. Director of the UC Berkeley FODA (Foundations of Data Analysis) Institute, Berkeley, CA, USA. Email: [mmahoney@stat.berkeley.edu](mailto:mmahoney@stat.berkeley.edu)

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